Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		In diamond each carbon atom is bonded to four other carbon atoms. (1) There are no free electrons to conduct electricity. (1) Whereas, in graphite each carbon atom is only bonded to three other atoms. (1) Therefore, there are delocalised (free) electrons in graphite, which can move and carry a charge. (1)	4			4		
		(ii)		There are only weak forces between layers (1) which allows layers of carbon to slide over each other in graphite. (1) On the other hand, in diamond all bonds are strong covalent bonds / there are no layers. (1)	3			3		
	(b)			Conversion of 0.36 nm into 3.6×10^{-10} m (1) Volume of cell = 4.7×10^{-29} m^3 (ecf)(1) Mass = $8 \times 2 \times 10^{-26}$ / 1.6×10^{-25} kg (1) Subs and answer = 3 400 (kg/m^3) (ecf) (1)		4		4	4	
	(c)			Carbon nanotubes have a much high(er) strength-to-weight ratio / high(est) strength <u>and</u> low(est) density (1) So will be strong<u>er</u> and light<u>er</u> (1)		2		2		
				Question 7 total	**7**	**6**		**13**	**4**	